Robot Control Systems **2025**

Lab 3: PID Position Control of the Qube-Servo 3

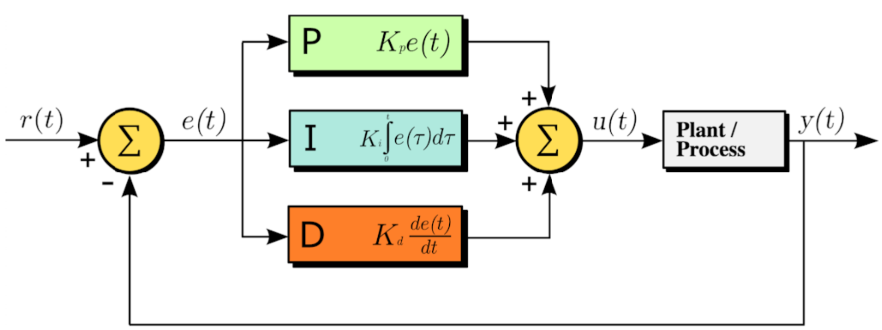
Purpose

Build on Lab 2's DC motor model to design and implement P/PD/PID controllers for *position* control. You will map time-domain specs to gains, validate in simulation, and test on the Qube (virtual $\rightarrow$ hardware).

Quick PID Recap

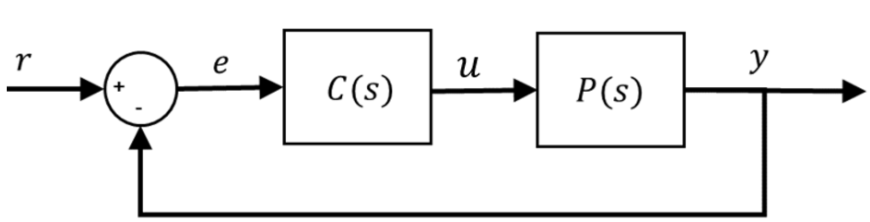
A PID controller uses $e(t) = r(t) - y(t)$:

$$u(t) = k_p\, e(t) + k_i \int_0^t e(\tau)\, d\tau \;+\; k_d \frac{de(t)}{dt}, \qquad C(s) = k_p + \frac{k_i}{s} + k_d s.$$



For unity feedback,

$$\frac{Y(s)}{R(s)} = \frac{C(s)P(s)}{1 + C(s)P(s)}.$$



On the Qube-Servo3 we will use:

$$P(s) = \frac{\Theta(s)}{V(s)} = \frac{K}{s(\tau s + 1)} \quad \text{from Lab 2 (or defaults } K{=}24,\ \tau{=}0.1).$$

Prelab

- Recall: standard 2nd-order form

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

with $\%OS = 100\, e^{-\pi\zeta/\sqrt{1-\zeta^2}}$, $t_p = \dfrac{\pi}{\omega_n\sqrt{1-\zeta^2}}$, $t_s \approx \dfrac{4}{\zeta\omega_n}$.

1 Tasks

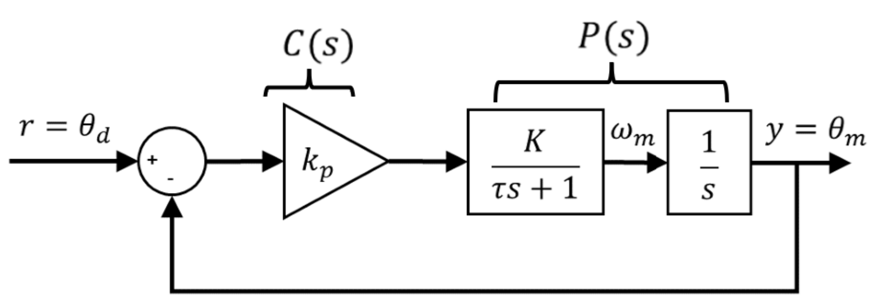
- Throughout, assume position control with $P(s) = \dfrac{K}{s(\tau s + 1)}$.

- Let $\Theta_m(s)$ and $\Omega_m(s)$ denote the Laplace transforms of angular position $\theta_m(t)$ and angular velocity $\omega_m(t)$, respectively.

- Similarly, let $\Theta_d(s)$ and $\Omega_d(s)$ denote the Laplace transforms of **desired** angular position $\theta_d(t)$ and angular velocity $\omega_d(t)$, respectively.

Task 1 — Proportional (P) control

Let $C(s) = k_p$.

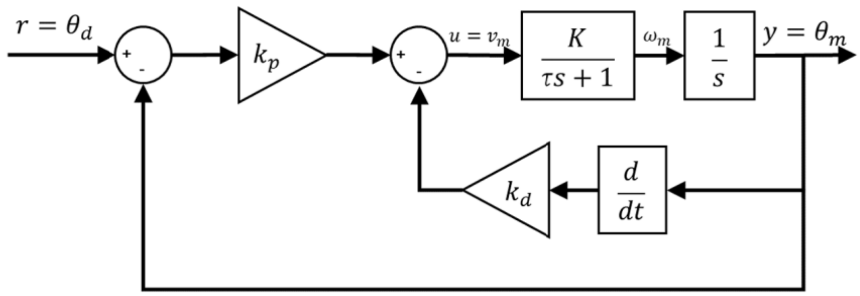


a) Derive $\dfrac{\Theta_m(s)}{\Theta_d(s)}$ (closed loop with unity feedback). *(Hint: use $P(s)$ above.)*

b) Discuss steady-state error to a step reference (will there be error? why?).

Task 2 — PD (rate feedback) control

Use the practical PD: $u = k_p(r - y) - k_d\,\dot{y}$ (derivative on measurement).



a) Derive the closed-loop transfer $\dfrac{\Theta_m(s)}{\Theta_d(s)}$.

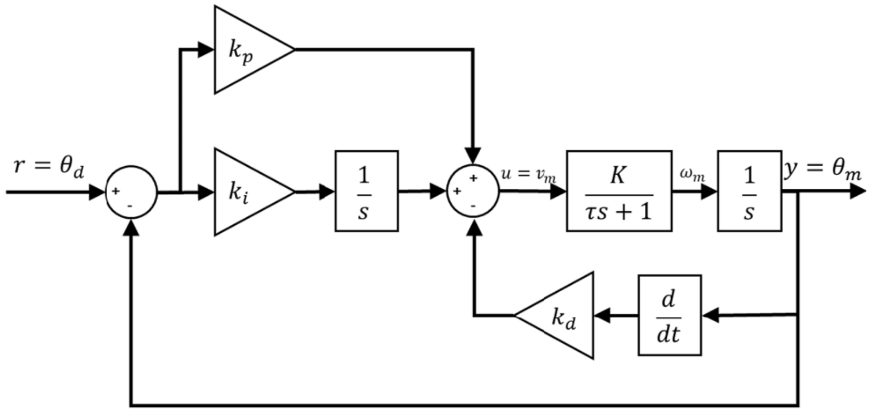
b) Match to $G(s) = \dfrac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$.

c) **Design from specs:** Given target $\%OS$ and t_p (or t_s), compute (ζ, ω_n), then k_p, k_d in terms of (K, τ).

Task 3 — PID (with derivative on measurement)

Use the modified PID:

$$u(t) = k_p e(t) + k_i \int_0^t e(\tau) d\tau \; - \; k_d \, \dot{y}(t).$$



a) Derive $\dfrac{\Omega_m(s)}{\Omega_d(s)}$ and show the cubic denominator:

Task 4 — Simulation & Qube test (saturation study)

a) **TF simulation (no saturation):** Implement $P(s)$ and your controller structure (P/PD/PID) where the derivative acts on the *measurement*. Plot the position step responses for a square-wave reference.

b) **Tighten specs:** Decrease $\%OS$ and t_p (i.e., larger ζ and ω_n). Recompute gains and simulate again. **Record the resulting overshoot** (estimate $\%OS$ from your data).

c) **Saturation toggle (software):** Remove saturation in software and observe TF predictions (they remain linear). Explain why TF does not reflect actuator limits.

d) **Re-enable saturation & Qube run:** For safety, *re-enable* saturation (e.g., ± 10 V). Run on the virtual device (then hardware if available). Compare measured $\%OS$, t_p, t_s, steady-state error, and peak $|u|$ to your TF predictions. Comment on discrepancies (deadband, friction b, sensor noise, saturation).